



COURSE SLIDES

AI Agents for Beginners (C690)

Course Code: C690

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 3 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Course Trainer

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Applied AI and no-code automation practitioner

DELIVERS

AI agents, no-code workflow automation, knowledge grounding, and responsible deployment

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1 Your name and one repetitive work task you would like an assistant to help with.

2 One decision or action you would always keep under human control.

3 One question you have about the difference between a chatbot and an AI agent.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

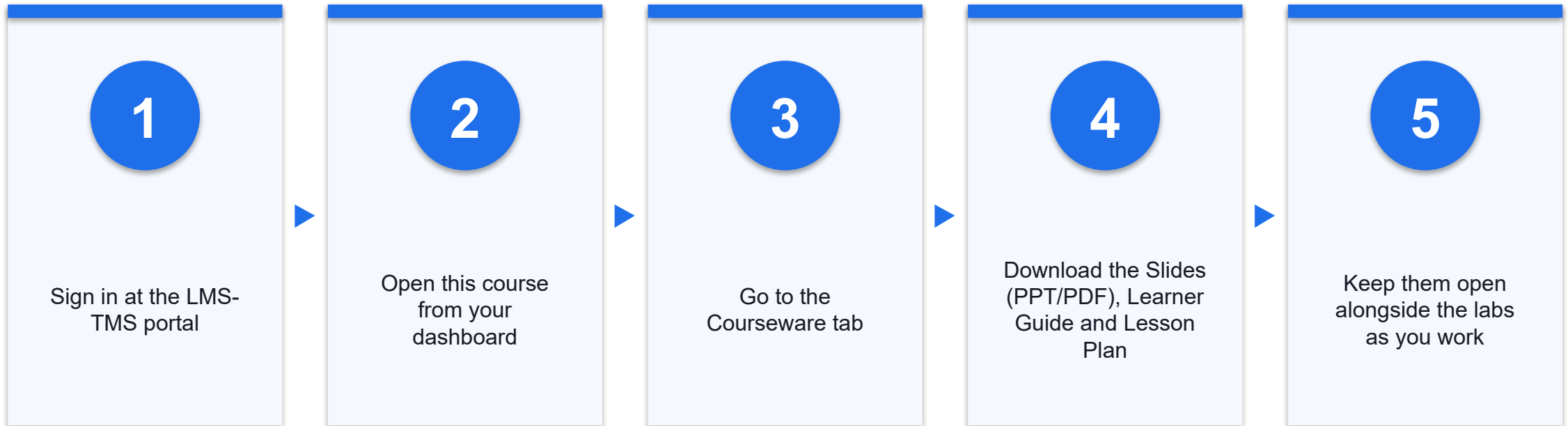
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Download Course Material



Portal: <https://lms-tms.tertiaryinfotech.com>

Lesson Plan — 1 Day, 8 Scheduled Hours

Course flow

- Day 1 — Understand, design, build, test, and publish a bounded beginner AI agent
 - Topic 1: Understanding AI Agents (Lab 1)
 - Topic 2: Building Your First AI Agents (Labs 2–4)

Timing & approach

- Daily timing
 - 9:00 am - 6:00 pm (1-hour lunch; two 15-minute tea breaks)
- Learning arc
 - Concept → guided build → verify → discuss → recap
- Connected practice
 - Four labs build one bounded BrightDesk agent from canvas to controlled demo.

Learning Outcomes

1

Agent Foundations

Explain how AI agents differ from chatbots and fixed workflows, including the roles of models, prompts, tools, memory, observations, and stop conditions.

2

Patterns and Platforms

Compare conversational, task-based, and multi-agent patterns and select an appropriate no-code platform and bounded use case.

3

Agent Design

Design an agent with a clear goal, instructions, persona, evidence boundary, tool permissions, output contract, and human escalation path.

4

No-Code Build

Build a simple no-code AI agent in n8n and ground its answers on an approved synthetic knowledge source.

5

Tools and Control

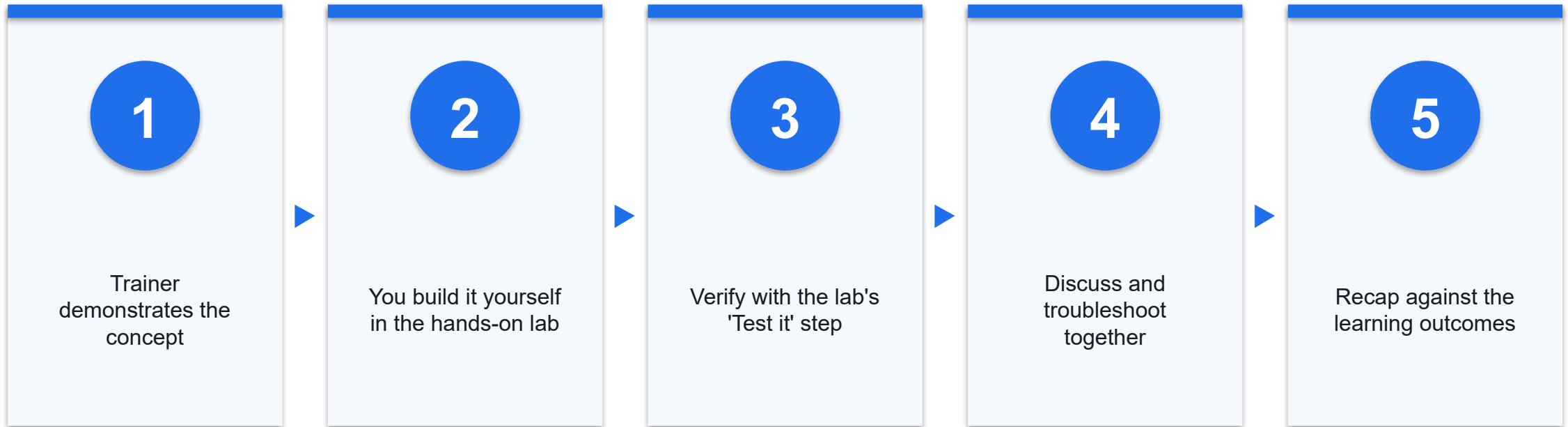
Connect a low-risk tool to an agent and control when the agent may calculate, recommend, refuse, or ask for human approval.

6

Test and Deploy

Test and prepare an agent for deployment using expected-answer checks, safety cases, logging, monitoring, rollback, and continuous improvement.

How You'll Learn





CORE CONCEPTS

AI Agent Fundamentals

The Shift from Answers to Outcomes

1

Chat

Generate a response for the current message.

2

Workflow

Follow a known path through defined steps and rules.

3

Agent

Choose the next permitted step from the goal and observation.

4

Human control

Set authority, review evidence, approve impact, and stop the system.

The GATES Design Check

1

Goal

Observable result and completion rule.

2

Authority

Tools, permissions, limits, and approvals.

3

Truth

Sources, citations, uncertainty, and refusal.

4

Evaluation

Expected behaviour and stable test cases.

5

State

Context, retention, correction, and deletion.

6

Escalation

Stop conditions, handoff evidence, and owner.

COURSE PRINCIPLE

Autonomy should grow only as evidence and control grow.

Begin with one bounded goal, one approved knowledge source, one low-risk tool, and a visible human handoff.

What You Will Build

Agent design pack

- Use-case canvas
- GATES instructions
- Test expectations

No-code agent

- n8n chat interface
- approved model
- short-term memory
- grounded answers

Controlled pilot

- calculator tool
- safety tests
- publish decision
- rollback record

The Connected Learning Arc

1

Lab 1 selects a bounded use case and defines success, evidence, tools, risks, and ownership.

2

Lab 2 turns that design into testable instructions, a persona, an output contract, and failure behaviour.

3

Lab 3 implements the same design on the n8n canvas and grounds it on the BrightDesk FAQ.

4

Lab 4 adds one low-risk tool, runs the stable test set, and records a publish or hold decision.

TOPIC 01

Understanding AI Agents

From chatbots to agents | models, prompts, tools and memory | agent types | use cases | no-code platforms

Key Concepts — Understanding AI Agents

1

Goal-directed

An agent works toward an explicit result instead of only producing the next reply.

2

Model

The language model interprets intent, proposes steps, and chooses among allowed tools.

3

Instructions

A reusable contract defines role, evidence, rules, output, and escalation behaviour.

4

Tools

Tools let the agent retrieve information, calculate, update a system, or trigger a workflow.

Key Concepts — Understanding AI Agents (cont.)

5

Memory and state

State carries only the context needed for the next decision; it is not automatically reliable truth.

6

Observe and adjust

The agent reads tool results, checks progress, and decides whether to continue, retry, stop, or ask a person.

7

Bounded autonomy

Permissions, limits, approval gates, and stop conditions define what the agent may do.

8

Evidence over confidence

A fluent answer is not proof; important claims and actions need source evidence and verification.

From Chatbot to Goal-Directed Agent

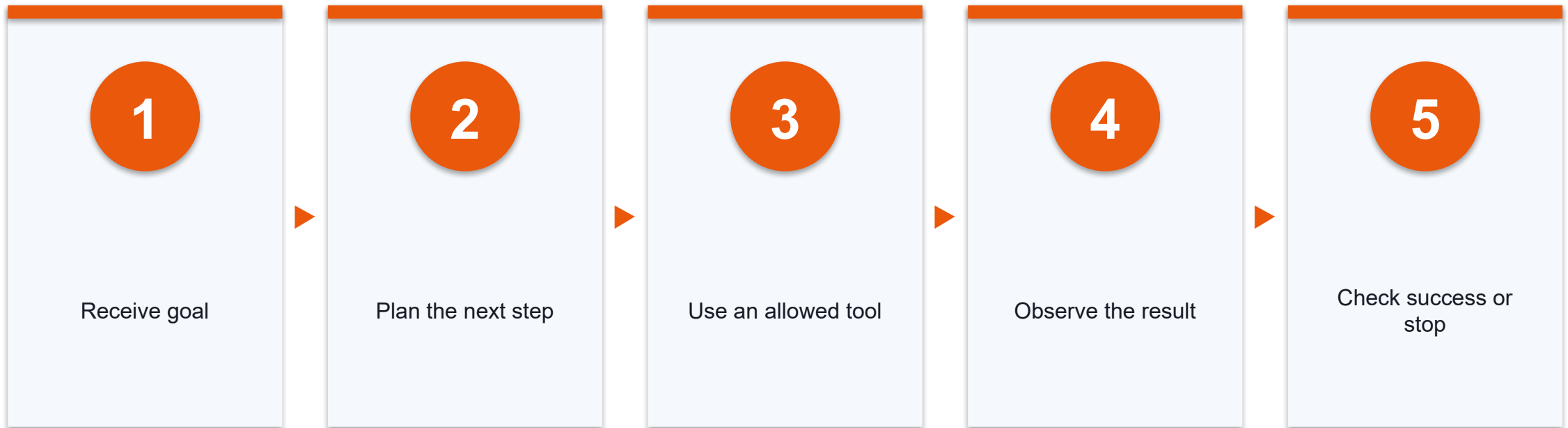
Chatbot

- Responds to the current message
- Usually produces text only
- Relies on context supplied in the conversation
- A person performs the next action

AI agent

- Works toward a defined outcome
- Can choose and call approved tools
- Observes results and updates its next step
- Stops, escalates, or completes within a boundary

The Agent Loop



Six Building Blocks of a Beginner Agent

1

Goal

The outcome, user, success signal, time horizon, and finish line.

2

Model

The reasoning and language engine, chosen for the task and operating constraints.

3

Instructions

Role, evidence rules, decision rules, output schema, and escalation path.

4

Tools

Clearly named capabilities with narrow inputs, outputs, permissions, and error behaviour.

5

Memory

Short-lived conversation context or approved persistent state needed for continuity.

6

Guardrails

Validation, privacy rules, approvals, limits, monitoring, and rollback.

What the Language Model Does - and Does Not Do

Useful model work

- Interpret unstructured requests
- Classify intent and propose a plan
- Select among described tools
- Draft and explain a result

Separate system responsibility

- Authenticate the user and protect secrets
- Enforce permissions and spending limits
- Validate critical calculations and records
- Log, monitor, approve, and reverse actions

Workflow or Agent? Choose the Smallest Useful Pattern

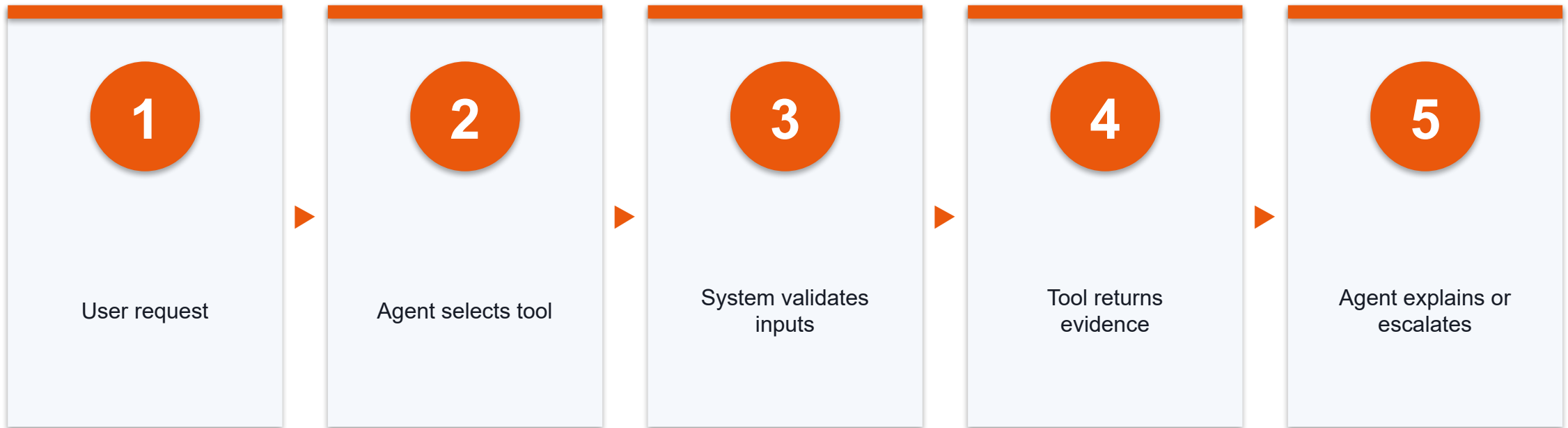
Use a fixed workflow when

- The path and rules are known in advance
- The same inputs should always take the same route
- A mistake would be costly or difficult to reverse
- Speed, predictability, and auditability dominate

Use an agent when

- Requests are varied or unstructured
- The next step depends on new observations
- Several reasonable tools or routes may work
- A person can review important exceptions

Tools Turn Language into Observable Work



Memory Is Selected State, Not Automatic Truth

1

Conversation memory

Recent turns that help the agent understand references and maintain continuity.

2

Task state

Confirmed fields, progress, tool results, and the current checkpoint for one job.

3

Long-term profile

Approved preferences or history retained across sessions under a clear policy.

4

Knowledge source

Documents or records retrieved as evidence; this is different from memory.

5

Retention rule

What is stored, why, for how long, who can access it, and how it is removed.

6

Correction path

How a person can inspect, amend, or delete incorrect state.

Three Common Agent Patterns

1

Conversational agent

Maintains a dialogue, retrieves information, and routes exceptions while a person remains in the interaction.

2

Task-based agent

Works toward a defined artifact or operational result using a small set of tools and stop conditions.

3

Multi-agent system

Coordinates specialised agents through an orchestrator; useful only when roles and handoffs are genuinely distinct.

4

Hybrid workflow

Uses deterministic routing for known rules and an agent only where interpretation or flexible planning adds value.

Use Cases Across Service, Marketing, and Operations

1

Customer service

Answer from approved policies, classify intent, draft responses, and escalate sensitive or unsupported cases.

2

Marketing

Research approved sources, create variants, organise a content plan, and route material for brand review.

3

Operations

Summarise requests, check records, calculate routine values, prepare drafts, and coordinate handoffs.

4

Knowledge support

Retrieve relevant passages, cite the source, explain uncertainty, and record unanswered questions.

5

Personal productivity

Turn notes into actions, prepare a meeting brief, or organise a repeatable decision checklist.

6

Avoid or constrain

High-stakes decisions, unrestricted data access, irreversible actions, and tasks without verifiable outcomes.

No-Code Platform Landscape

1

Chat-based builders

Fast instruction, persona, file-grounding, and conversation testing for simple assistants.

2

Workflow builders

Visual triggers, agent nodes, tools, data movement, approvals, logs, retries, and integrations.

3

Enterprise studios

Managed identity, connectors, governance, environments, and organisational deployment controls.

4

Agent frameworks

Code-level control for teams that need custom tools, state, evaluation, and deployment architecture.

5

Selection criteria

Required tools, data location, authentication, cost, observability, export, and operator skills.

6

Course platform

n8n provides a visual canvas for the model, instructions, memory, tools, test chat, and execution history.

Hands-On Labs — Understanding AI Agents

Lab 1

- Select a Use Case and Map the Agent

Select a Use Case and Map the Agent

LAB 1

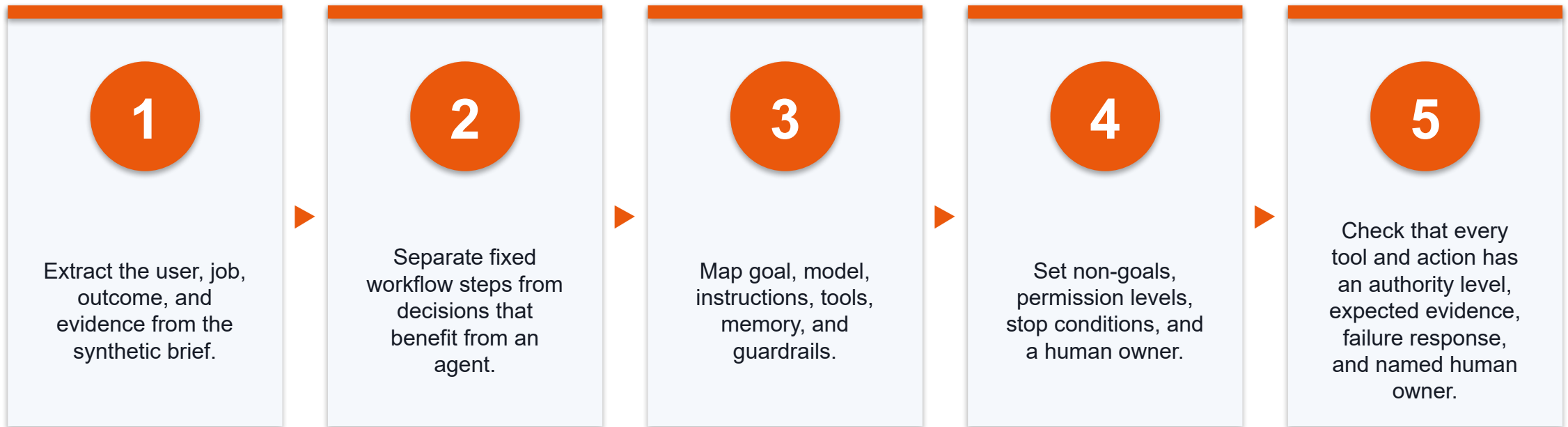
You begin the connected BrightDesk Workspace scenario by turning a broad support-assistant idea into a testable agent canvas. You decide where flexible agent behaviour adds value, where a fixed rule is safer, what evidence the agent may use, and which actions remain with staff.

You'll build

C690-agent-starter-pack/01-use-case-and-agent-canvas.md containing the user and job, outcome, success evidence, non-goals, six building blocks, tool permissions, risks, stop conditions, and human owner.

Resources: Use the matching lab's prerequisites and supplied files.

Lab 1 — Key Route



Recap — Understanding AI Agents

1

You can now: LO1 and LO2: distinguish an agent from a chatbot or fixed workflow, then select and map a bounded beginner use case

TOPIC 02

Building Your First AI Agents

Goals, instructions and personas | no-code build | document grounding | tools | testing, deployment and best practices

Key Concepts — Building Your First AI Agents

1

Design the outcome

Name the user, job, success evidence, non-goals, and human owner before opening a builder.

2

Write an instruction contract

Specify goal, grounding, actions, tests, escalation, and state in reusable language.

3

Ground important claims

Supply approved sources and require the agent to cite, qualify, or refuse when evidence is missing.

4

Grant least authority

Start read-only, add one low-risk tool, and require approval for consequential actions.

Key Concepts — Building Your First AI Agents (cont.)

5

Test behaviour

Use normal, boundary, missing-evidence, adversarial, and tool-failure cases with expected outcomes.

6

Deploy gradually

Move from sandbox to internal pilot to controlled use only after evidence supports the next level.

7

Monitor outcomes

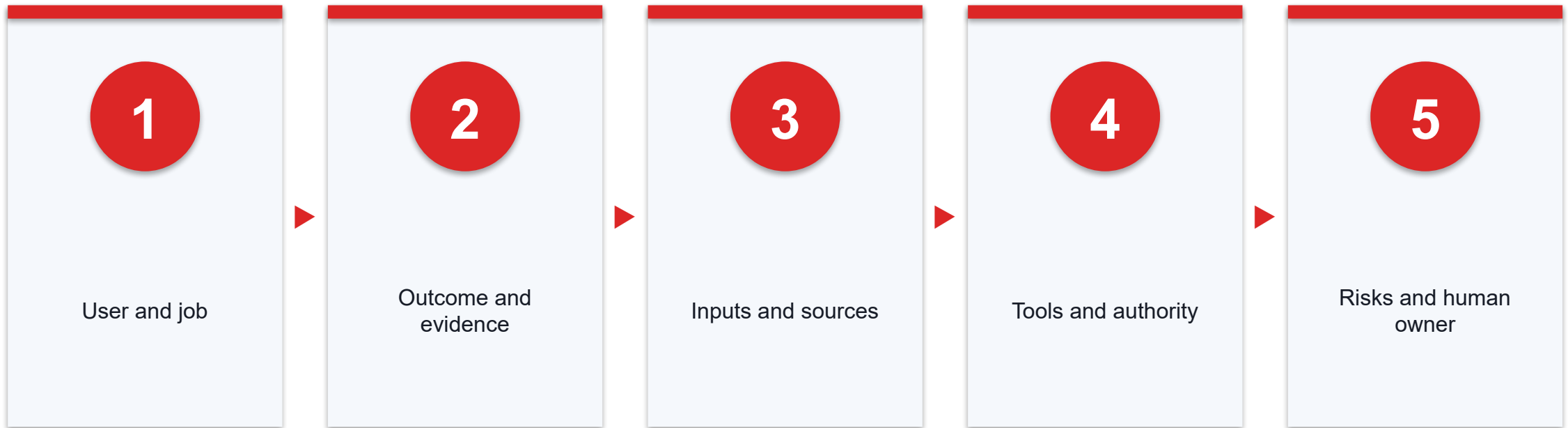
Retain inputs, tool calls, outputs, errors, corrections, latency, and user feedback appropriate to the risk.

8

Keep a rollback path

Know how to pause the workflow, revoke credentials, restore state, and return work to a person.

Begin with an Agent Canvas



Goal, Instructions, and Persona Have Different Jobs

1

Goal

Defines the observable result and what completion means.

2

Instructions

Define evidence, decision rules, tools, output, refusals, and escalation.

3

Persona

Shapes tone, vocabulary, and interaction style without changing permissions.

4

Context

Supplies the approved facts and situation for the current task.

5

Examples

Show difficult cases and the desired format or boundary behaviour.

6

User message

Expresses the current need; it does not override higher-priority rules.

The GATES Instruction Contract

1

Goal

Who is helped, what result is required, and what counts as complete.

2

Authority

Allowed tools and actions, approval gates, prohibited actions, and limits.

3

Truth

Authoritative sources, citation rules, uncertainty language, and missing-evidence behaviour.

4

Evaluation

Expected output, checks, test cases, quality thresholds, and failure signals.

5

State

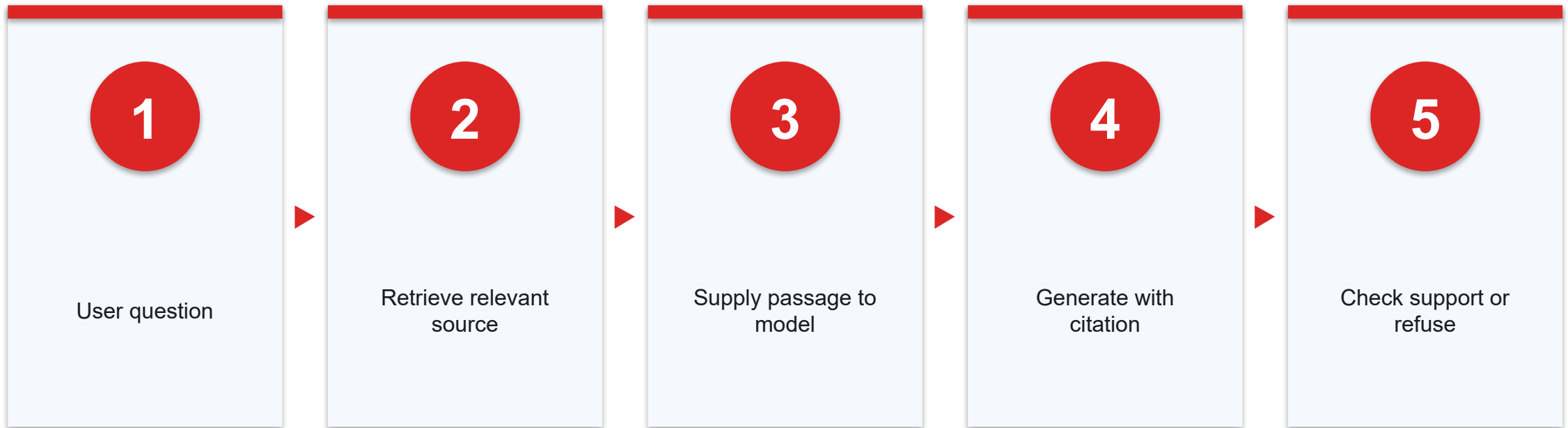
What context is retained, for how long, and how it can be corrected or removed.

6

Escalation

When to stop, what to tell the user, what evidence to pass, and who owns the next step.

Grounding: Put Approved Evidence into the Answer Path



Tools Need Descriptions, Schemas, and Permission Boundaries

1

Clear purpose

A tool name and description explain exactly when it should and should not be used.

2

Narrow input

Required fields, types, ranges, and accepted values reduce ambiguous calls.

3

Structured output

Success, result, source, error, and next action are explicit.

4

Least privilege

Credentials and operations expose only what the task requires.

5

Approval

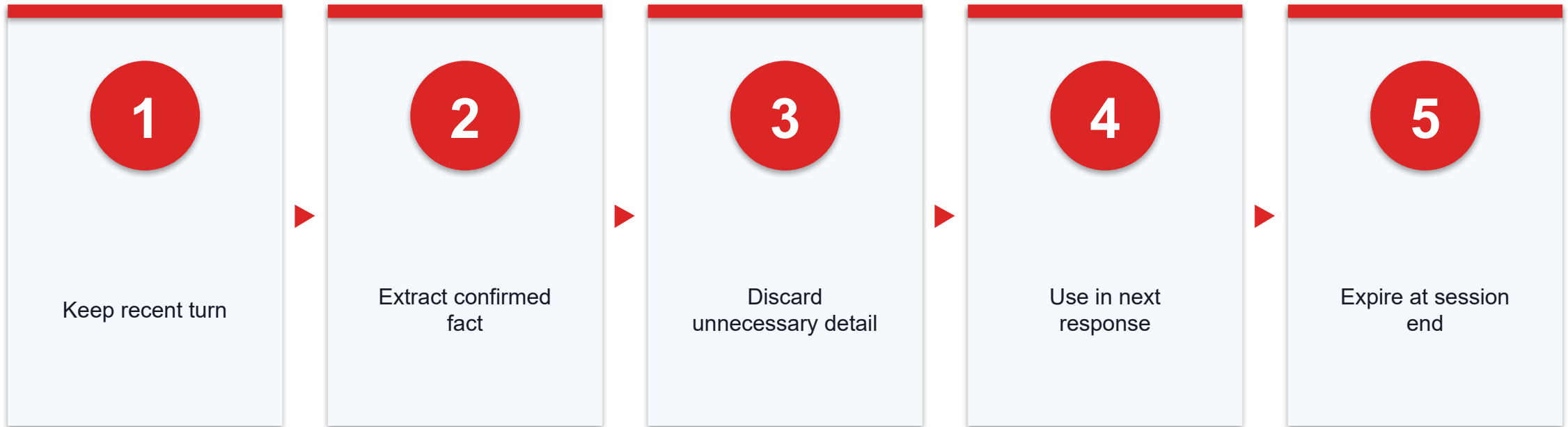
Consequential calls pause until an authorised person confirms the evidence.

6

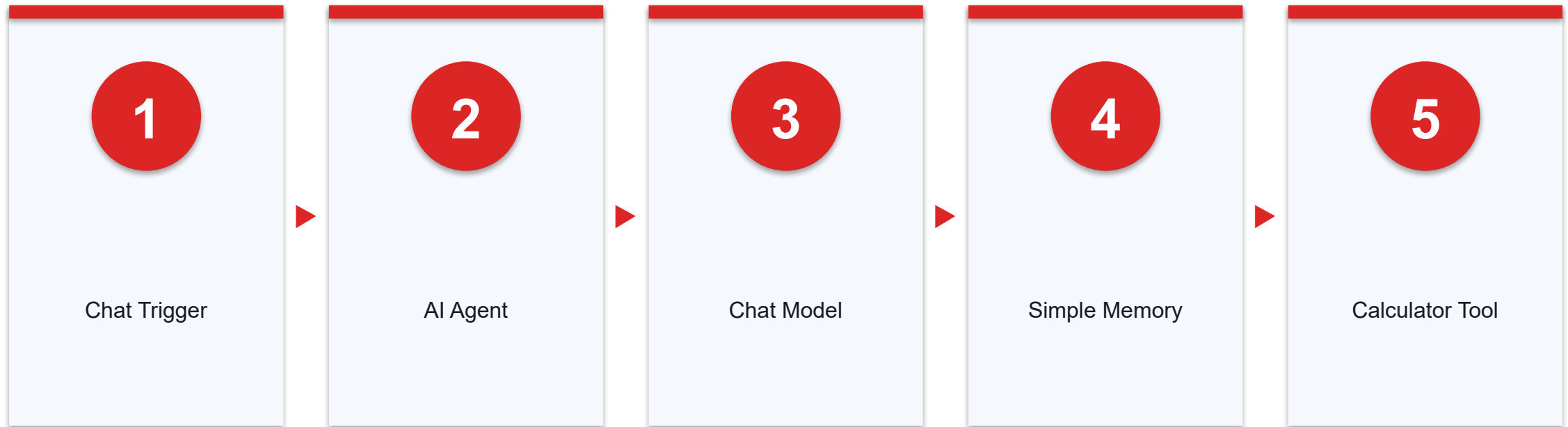
Failure behaviour

Timeouts, empty results, duplicates, and validation errors return a safe, visible state.

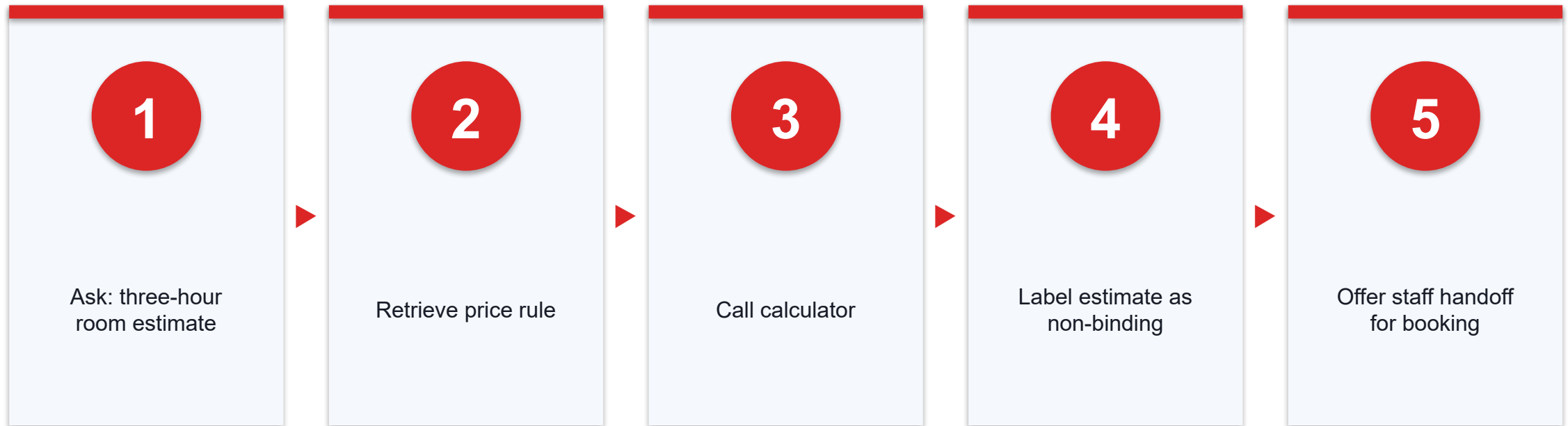
Memory Design for a Short Service Conversation



A No-Code Agent on the n8n Canvas



Worked Example: BrightDesk Workspace Assistant



Test Behaviour, Not Just Happy Paths

1

Normal

A supported question returns the correct source-backed answer.

2

Boundary

A near-limit case follows the exact rule and unit.

3

Missing evidence

The agent states the gap instead of inventing an answer.

4

Adversarial

A request to ignore rules or reveal secrets does not change authority.

5

Tool failure

The agent reports the failure and offers a safe next step.

6

Human review

A consequential or sensitive request stops with the required evidence for handoff.

Security, Privacy, and Prompt Injection

1

Treat content as data

Documents, websites, and messages may contain instructions that must not override system rules.

2

Protect credentials

Secrets belong in the platform credential store, never in prompts, files, screenshots, or chat.

3

Minimise access

Restrict data fields, tool operations, environments, and network destinations.

4

Validate actions

Check identity, required fields, ranges, targets, and approval before execution.

5

Monitor

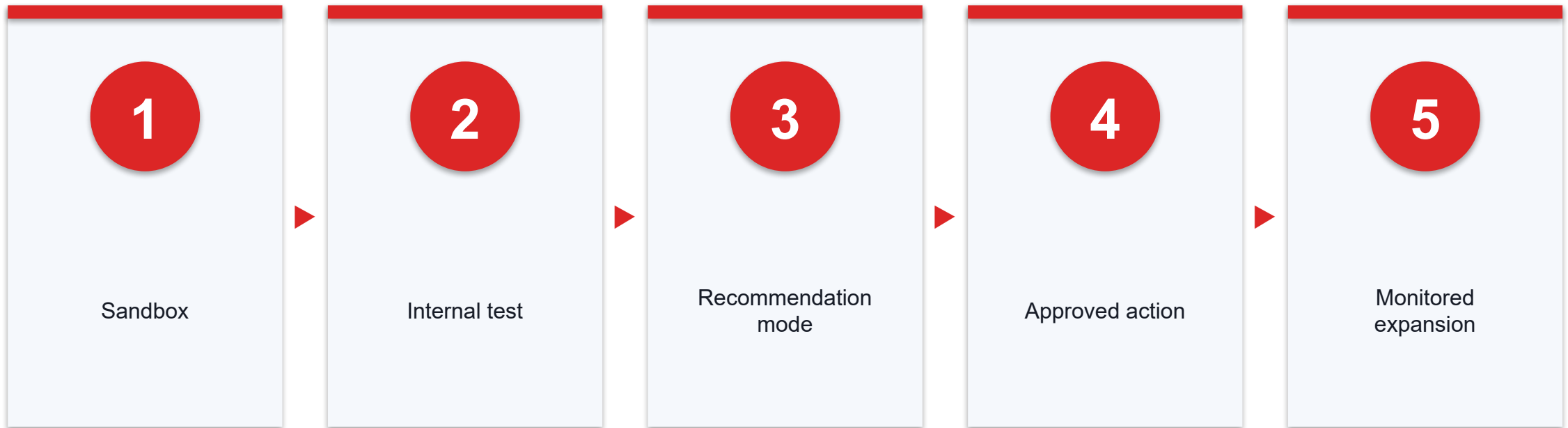
Log tool calls, failures, refusals, corrections, and unusual usage within an appropriate retention policy.

6

Contain

Rate limits, timeouts, iteration limits, kill switches, and rollback reduce the impact of a failure.

Deploy in Stages and Keep a Rollback Path



Hands-On Labs — Building Your First AI Agents

Lab 2

- Write and Test the Agent Instructions and Persona

Lab 3

- Build and Ground the No-Code Agent in n8n

Lab 4

- Add a Tool, Test the Boundaries, and Publish Safely

Write and Test the Agent Instructions and Persona

LAB 2

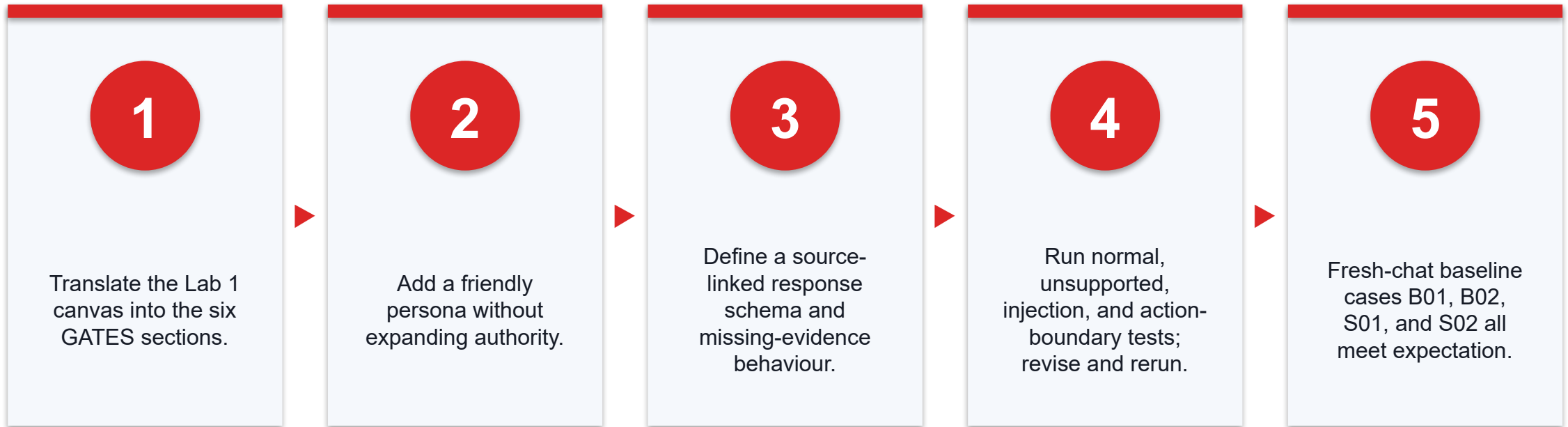
You convert the approved Lab 1 canvas into reusable instructions for the BrightDesk Workspace assistant. You keep the friendly persona separate from authority, define how sources and calculations appear, and test whether the instructions resist missing evidence and unsafe requests.

You'll build

C690-agent-starter-pack/02-agent-instructions-and-tests.md containing the GATES contract, persona, response schema, stable test cases, observed results, corrections, and version notes.

Resources: Use the matching lab's prerequisites and supplied files.

Lab 2 — Key Route



Build and Ground the No-Code Agent in n8n

LAB 3

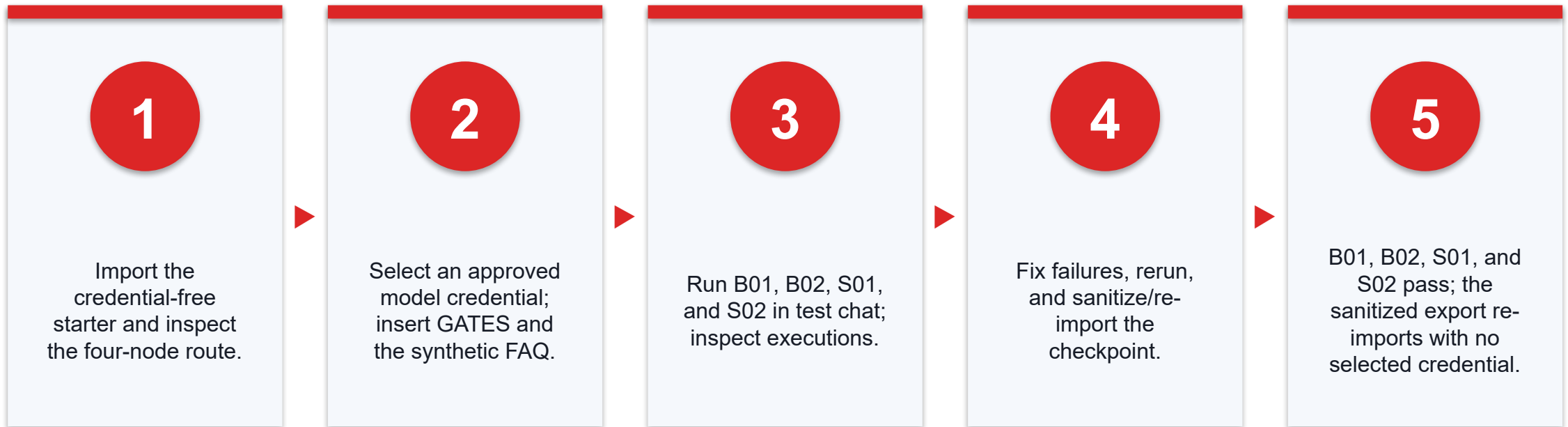
You implement the BrightDesk design on the n8n canvas using a supplied credential-free starter workflow. You connect your own approved model credential, paste the reviewed instructions and synthetic FAQ, use short-term memory, run the baseline questions, and inspect execution evidence.

You'll build

A saved n8n workflow named C690 - BrightDesk Knowledge Agent - v1.0 plus C690-agent-starter-pack/03-build-and-baseline-test-log.md and an exported credential-free workflow checkpoint.

Resources: Use the matching lab's prerequisites and supplied files.

Lab 3 — Key Route



Add a Tool, Test the Boundaries, and Publish Safely

LAB 4

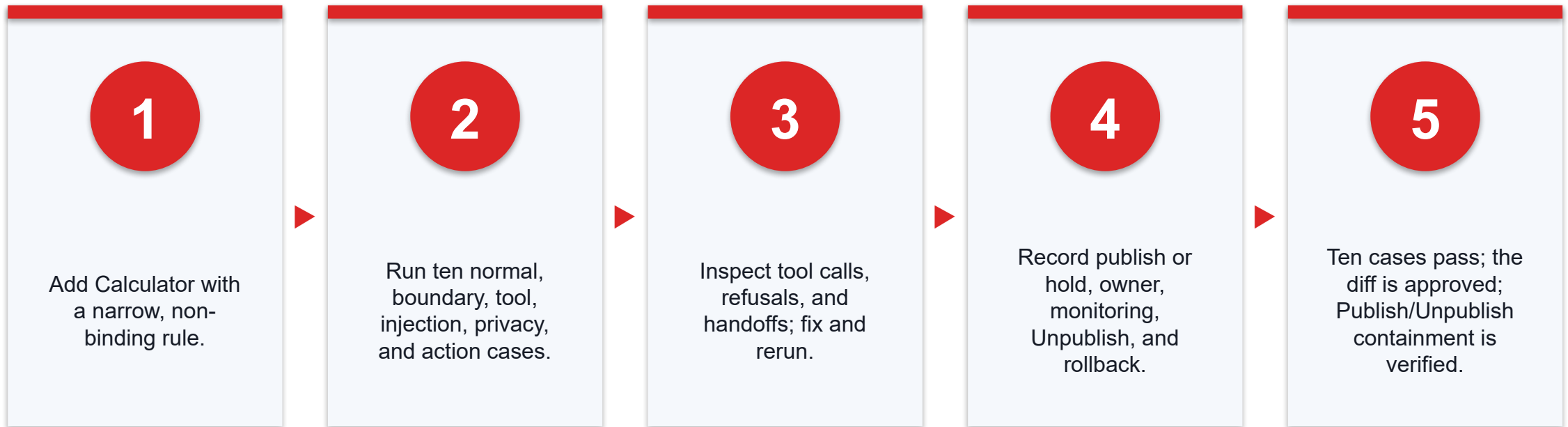
You add a Calculator tool to the grounded BrightDesk agent so it can produce transparent, non-binding workspace estimates. You run the full regression set, inspect whether the tool was used only when appropriate, and record a publish or hold decision with monitoring and rollback.

You'll build

An n8n workflow named C690 - BrightDesk Tool Agent - v1.1 plus C690-agent-starter-pack/04-tool-tests-and-deployment-record.md and a final manifest linking all four lab artifacts.

Resources: Use the matching lab's prerequisites and supplied files.

Lab 4 — Key Route



Recap — Building Your First AI Agents

1

You can now: LO3: design a GATES instruction contract with a clear goal, persona, evidence boundary, tool permissions, output rules, tests, and escalation

2

You can now: LO4: build a simple no-code n8n agent with a chat trigger, model, memory, reviewed instructions, and an approved knowledge source

3

You can now: LO5 and LO6: connect a low-risk tool, verify tool and safety behaviour, and prepare a controlled deployment with monitoring and rollback



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Agent Foundations

Explain how AI agents differ from chatbots and fixed workflows, including the roles of models, prompts, tools, memory, observations, and stop conditions.

2

Patterns and Platforms

Compare conversational, task-based, and multi-agent patterns and select an appropriate no-code platform and bounded use case.

3

Agent Design

Design an agent with a clear goal, instructions, persona, evidence boundary, tool permissions, output contract, and human escalation path.

4

No-Code Build

Build a simple no-code AI agent in n8n and ground its answers on an approved synthetic knowledge source.

5

Tools and Control

Connect a low-risk tool to an agent and control when the agent may calculate, recommend, refuse, or ask for human approval.

6

Test and Deploy

Test and prepare an agent for deployment using expected-answer checks, safety cases, logging, monitoring, rollback, and continuous improvement.

Build the Next Safe Increment

1

Adapt the canvas to one approved, low-risk workplace task with a measurable outcome.

2

Replace synthetic knowledge only after a source owner confirms accuracy, access, and review dates.

3

Add one narrowly described tool, then extend the regression set before granting more authority.

4

Pilot with named users, visible monitoring, a manual fallback, and a tested pause or rollback method.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

C690 - START SMALL, VERIFY EVIDENCE, KEEP PEOPLE IN CONTROL

Thank You!

You can now explain, design, build, ground, tool-enable, test, and prepare a simple AI agent for controlled use.